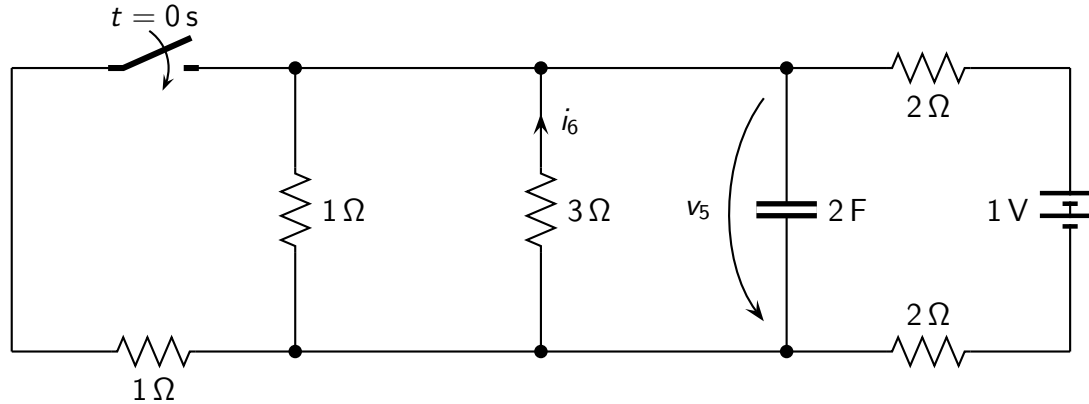


Problem: find the expression of $v_5(t)$, $i_6(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{24}{31} \text{ s}$;
- Initial conditions before switching: $v_5(0^-) = \frac{-3}{19} \text{ V}$;
- Initial conditions after switching: $v_5(0^+) = \frac{-3}{19} \text{ V}$; $i_6(0^+) = \frac{-1}{19} \text{ A}$;
- Steady-state solution: $v_5(\infty) = \frac{-3}{31} \text{ V}$; $i_6(\infty) = \frac{-1}{31} \text{ A}$;
- Solution for $t > 0$:

$$v_5(t) = \left(-61.121 \cdot 10^{-3} e^{-t/\tau} + \frac{-3}{31} \right) \text{ V}$$

$$i_6(t) = \left(-20.374 \cdot 10^{-3} e^{-t/\tau} + \frac{-1}{31} \right) \text{ A}$$